

GreenStep Employee Sustainability Challenge

Product Requirements Document

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1 Product Overview

GreenStep Employee is a web-based sustainability challenge platform that helps employees log eco-friendly actions, earn points and badges, participate in teams, and compare progress through leaderboards. The product replaces spreadsheet-based tracking with a centralized, low-maintenance system that is more engaging for end users and easier to manage for administrators.

The primary user groups are:

- **Employees**, who join challenges, log actions, and monitor their own progress.
- **Administrators**, who configure challenges, review team requests, and export participation data.

2 Core Functional Scope

At the end of the Spring 2026 capstone, the product supports:

- secure sign-in and protected dashboard access,
- joining challenges through a join code,
- joining or leaving a team within a challenge,
- submitting team requests for administrator review,
- logging sustainability actions across multiple categories,
- awarding points, badges, streaks, and leaderboard positions,
- viewing challenge-specific and global activity data,
- category and challenge filtering for dashboard insights, and
- exporting CSV data from the admin workspace.

3 Detailed Use Cases

3.1 Use Case 1: Employee Logs a Sustainability Action

Primary Actor: Employee

Goal: Record a completed sustainability action and receive the appropriate points inside the correct challenge context.

Preconditions:

- The employee is authenticated.
- The employee has completed onboarding.
- The employee is on the dashboard or log action page.

Trigger: The employee decides to log a completed action such as biking to work, recycling plastic, or turning off unused lights.

Main Success Scenario:

1. The employee opens the **Log Action** experience.
2. The employee browses by category, recent actions, or popular actions, or uses search to find a relevant action.
3. The employee selects an action from the list.
4. The system opens the action detail page and shows the base point value.
5. The employee optionally adds a story or image.
6. The employee submits the action.
7. The system validates the request and checks for duplicate rapid submissions.
8. The system stores the action in the database, scoped to the selected challenge or the global activity context.
9. The system updates the employee's points, activity history, streak, badges, and relevant leaderboard views.
10. The system returns the employee to a success state where the updated action is visible in activity feeds and dashboard summaries.

Alternative / Exception Flows:

- If the employee submits the same action too quickly, the system rejects the duplicate submission and shows an error.
- If the employee is no longer authenticated, the system blocks the action and requires sign-in.

- If an optional image or note is omitted, the action is still accepted with only the applicable points.

Postconditions:

- A new action record exists in the database.
- The employee's activity feed and points reflect the logged action.
- Challenge-specific and global views remain consistent with the new data.

3.2 Use Case 2: Administrator Reviews a Team Request and Manages Challenges

Primary Actor: Administrator

Goal: Manage challenge participation by reviewing pending team requests and configuring challenge data from the admin workspace.

Preconditions:

- The administrator is authenticated.
- The administrator's Clerk metadata role is set to **admin**.

Trigger: The administrator opens the admin workspace to review a team request or create/export challenge data.

Main Success Scenario:

1. The administrator navigates to the admin page.
2. The system verifies that the current user has admin access.
3. The system displays pending team requests, recently reviewed requests, challenge summaries, and export controls.
4. The administrator opens a pending team request.
5. The administrator reviews the requested team name, requester, and challenge context.
6. The administrator chooses to approve or reject the request.
7. If approved, the system creates the team if needed, adds the user to the team, and updates the request status.
8. If rejected, the system records the rejection status and reviewer.
9. The administrator may also create a new challenge, generate demo challenges, or export activity data to CSV.
10. The system revalidates the affected dashboard and admin views so the updated state is immediately reflected.

Alternative / Exception Flows:

- If a non-admin user attempts to access the admin workspace, the system blocks access and displays an admin-only message.
- If a team request is invalid or has already been processed, the system rejects the action and preserves data integrity.
- If the administrator attempts to create a duplicate team name within the same challenge, the system prevents the duplicate.

Postconditions:

- The request is stored with an approved or rejected status.
- Relevant team, challenge, and dashboard data remain synchronized.
- Administrators retain a recent history of reviewed requests.

4 User Stories with Acceptance Criteria

4.1 User Story 1: Join a Challenge

Story: As an employee, I want to join a challenge with a code so that my actions and team participation are associated with the correct campaign.

Acceptance Criteria:

- The user can open a join challenge modal from the dashboard.
- The user can enter a challenge code and submit it.
- If the code is valid and the challenge is active, the user is added to that challenge.
- If the user already joined the challenge, the system prevents a duplicate join and shows an explanatory error.
- If the challenge has ended or the code is invalid, the system displays a clear error message.

4.2 User Story 2: View Filtered Progress by Challenge or Global Scope

Story: As an employee, I want to switch between all activity and a specific challenge so that I can understand my performance in the right context.

Acceptance Criteria:

- The dashboard provides a scope selector for all activity and joined challenges.
- When the user switches scope, points, leaderboard data, and activity summaries update to match that scope.
- Global views aggregate lifetime data across the system.
- Challenge views show only challenge-specific points, actions, and team context.

- The system preserves consistent labels so the user can distinguish global and challenge-scoped information.

4.3 User Story 3: Submit a Team Request

Story: As an employee without a team, I want to request a new team in my challenge so that an administrator can review and approve it.

Acceptance Criteria:

- The user can open a request form from the team area of the dashboard.
- The user must already belong to the challenge before submitting a request.
- The request captures the desired team name and optional message.
- The system prevents duplicate pending requests for the same context.
- Submitted requests appear in the user's request history and in the admin review queue.

4.4 User Story 4: Export Participation Data

Story: As an administrator, I want to export sustainability activity data so that I can analyze participation outside the application and share results with stakeholders.

Acceptance Criteria:

- The admin workspace provides CSV export controls.
- The administrator can export all activity or challenge-specific activity.
- The generated CSV includes enough participation data to support reporting and review.
- Only authorized administrators can access export actions.
- The exported data reflects the latest saved records in the system.

5 Notes for Future Teams

Future teams should treat action logging, challenge membership, team membership, and admin request review as tightly connected product areas. Changes in one area often affect leaderboards, dashboard summaries, and export behavior. Before extending the system, teams should confirm whether a feature is global, challenge-scoped, or admin-only and keep that distinction explicit in both the UI and API design.